Vorlesung
Informationsvisualisierung

KLAUSUR-VORBEREITUNG



Lecture
Information Visualization

EXAM PREPARATION

Prof. Dr. Stephan Diehl

Universität Trier – Sommersemester 2026

Final Exam



- Thursday, 16.07.2026 in **HS13** at 12:**00** **(We will also use K101)**
- be there on time please
- bring along student ID
- duration: 90 minutes
- No auxiliary material may be used!
- Please write with black or blue ink. **Lead pencils are not permitted!**
- Other colors are only allowed for design questions
- Exam questions
  - Knowledge
  - Understanding
  - Transfer







Exam Questions

- Knowledge:
  - Name four …
  - Give the definition of …
- Understanding:
  - Explain …
  - Explain how technique/algorithm X works".
  - Explain or fix program code
  - Compare technique A to technique B (e.g. advantages, disadvantages)
- Transfer:
  - Apply a technique to a given data set (e.g. draw a certain kind of diagram for a given set of values)
  - Design a visualization for a given task and a given data set
  - Program code or snippet (in Java or Python) for a small visualization task

Design Task from Sample Exam

Question 3 For a book that was co-authored by several authors (up to 10) we have the following change data: what was changed (chapter, section), length of change (number of words), who did the change (author name) and when was the change done (timestamp). Design a visualization approach that helps to explore this data and to detect groups of authors who work together as well as patterns on how they work together. **Explain** your visualization concept, draw sketches to **illustrate** your approach and **justify** your design decisions.

List of tools or diagrams you should know.

Diagrams

- Line chart
- Bar chart
- Pie chart
- Sparklines
- Pictograms
- IsoTypes

- Jackson diagram
- Nassi-Shneidermann diagram
- Control-flow diagram

Hierarchy Visualization

- Node-link diagrams
  - Radial trees
  - Cone Trees
  - Botanical visualization of trees
- Indented outline plot
- Icicle plot
- Sunburst
- Treemap
- Cushion Treemap
- Information Pyramids

Graph visualization

- Node-link diagrams
  - Radial / force-directed / hierarchical / orthogonal layout
  - Edge bundling
- Matrix visualization
  - Nodetrix
- Dynamic graphs
  - Animation
  - Timeline

Multivariate Data

- Minard's drawing
- Boxplots
- Fan-Chart
- Histogram
- Extended Table
- Table Lens
- Scatterplot
- Parallel Coordinates
- Star Plot

Time Series

- Interactive Spiral

List of tools or diagrams that could inspire your own design ideas (by analogy)

- Code visualization
  - Control-structure diagram
  - Code Canvas
  - Code Bubbles
- Software architecture
  - UML diagrams (not in detail)
  - Geon diagrams
  - Class blueprint
  - Dependency structure matrix
- Algorithm animation
  - Sorting out sorting
  - Samba
- Visual Debugging
  - DDD
  - Reference patterns
  - Tarantula
- Software evolution
  - SeeSoft
  - CVSscan
  - Evolution Matrix
  - EPOSee
  - Evolution Radar